SOFC PLANT · HOT STANDBY · STACK VOLTAGE

Stack voltage in hot standby: steady, but one stack is sagging

In hot standby each string of 4 stacks shows about **420–475 V**, roughly **110 V per stack**. No current is drawn, so this is the resting voltage, like a battery's voltage with nothing plugged in. It holds steady through a run and follows the hydrogen supply. One stack (string 4, stack 3) sits well below the rest, and it got worse over March.

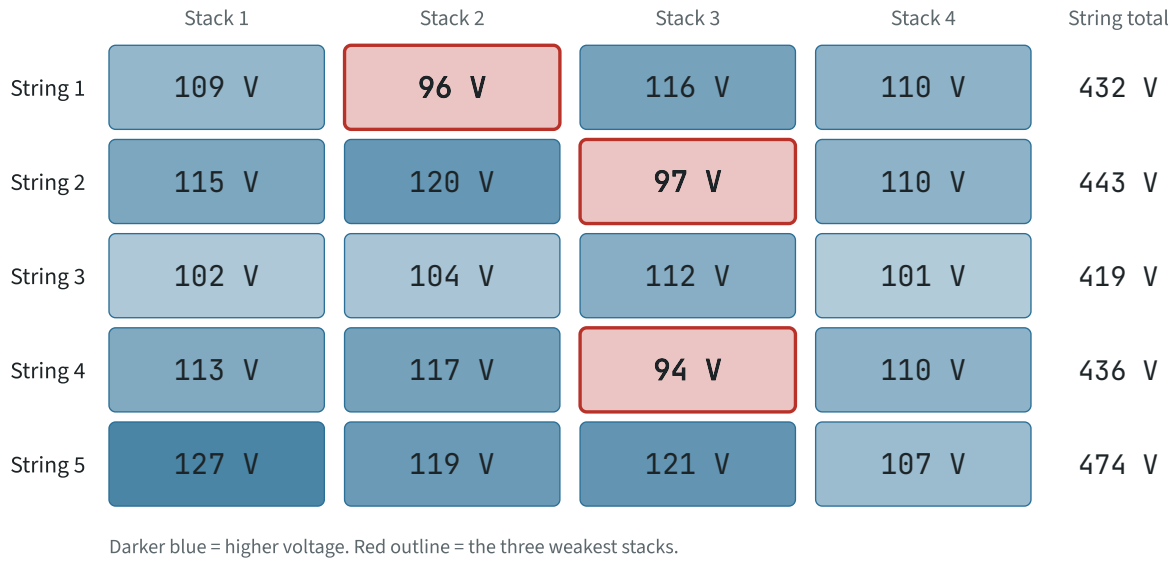
Why it matters: with no current flowing, every stack gets the same gas and the same heat. A stack that rests lower than its neighbours is the earliest sign that something is different inside it. Hot standby is the cleanest place to spot that.

~110 V
typical stack, settled hot standby

419–474 V
string totals (median per string)

−7.5 V
per stack for every 100 °C hotter

94 V
stack 4.3, lowest of the 20 (about 85 V by late March)



Darker blue = higher voltage. Red outline = the three weakest stacks.

Median voltage of each of the 20 stacks over all settled hot-standby minutes (first 30 min of each run left out). 24 hot-standby runs, 338 hours, 31 Jan – 30 Mar 2026. Name "4.3" means string 4, stack 3.

What we checked

QUESTION	ANSWER	PAGE
How high is the voltage?	~110 V per stack, 419–474 V per string. String 5 is always highest, string 3 lowest.	this page
Does it move during a run?	Flat once settled. Drops when hydrogen is cut, rises when natural gas comes in.	2
Is any stack unusual?	Stack 4.3 is the lowest, and it fell from ~90 % to ~80 % of its neighbours over March.	3
Can we trust the numbers?	Mostly. The "minimum cell voltage" tag is not a real cell, and three stack readings dropped out for 2 days in February.	4

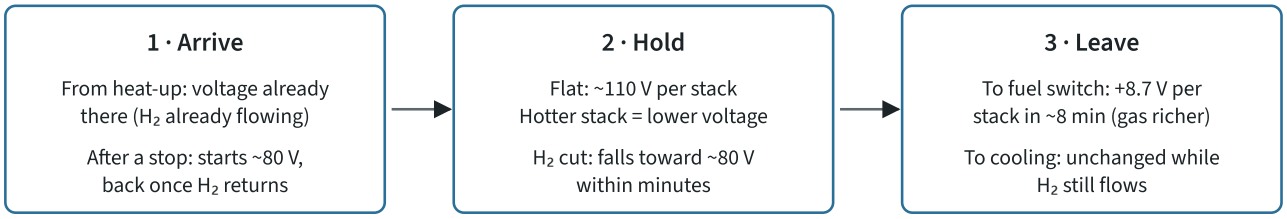
> **IF YOU REMEMBER ONE THING**
> Hot-standby voltage is steady at about 110 V per stack, except stack 4.3, which rests about 20 V lower than its neighbours and is getting weaker.

THROUGH ONE STANDBY RUN →

PAGE 2 · THROUGH ONE STANDBY RUN

The voltage follows the gas, then sits still

In hot standby the stack gets a small, steady stream of hydrogen (60 litres a minute in 1,200 of nitrogen). That trickle sets the voltage. While it flows, the voltage is flat. When it stops, the voltage falls. When natural gas arrives, the voltage jumps up.



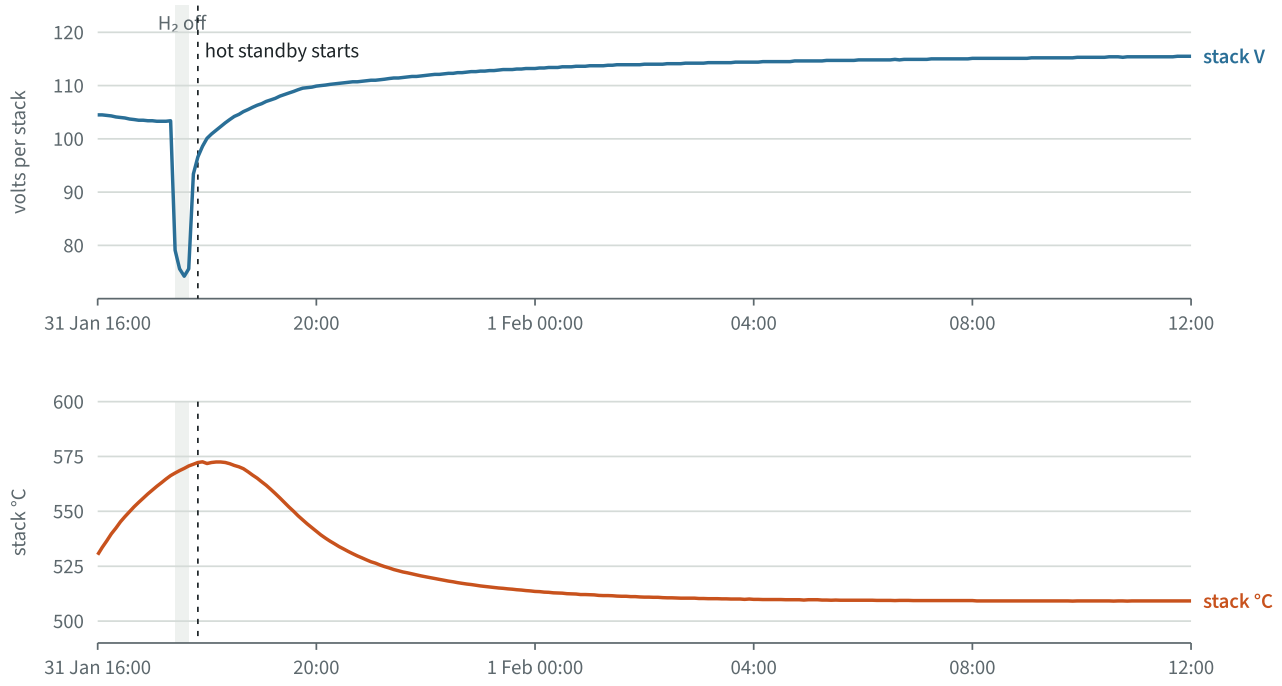
No current in any step: the voltage is set by gas and temperature only

PLAY THE THREE STEPS

What the stack voltage does when it arrives in hot standby, while it holds there, and when it leaves. Medians over 24 runs.

Hold: flat, and slightly lower when hotter

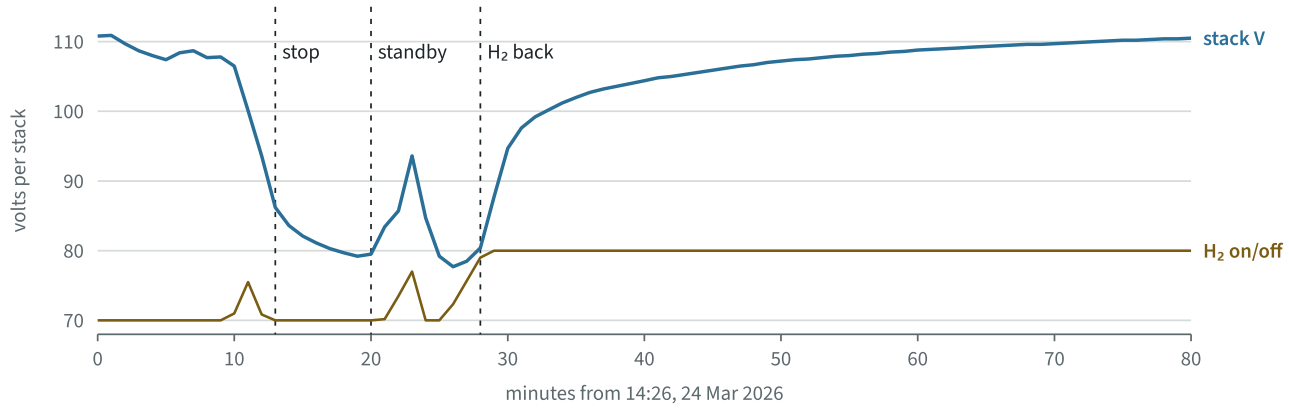
The first hot-standby run is the longest clean example. The stack cools from 572 to 509 °C and the voltage climbs from 102 to 115 V per stack, then levels off. Across all runs, a stack loses about **7.5 V for every 100 °C** it is hotter (≈ 30 V per string). So a run held at 600 °C rests a few volts lower than one held at 510 °C. That is normal chemistry, not a fault.



31 Jan – 1 Feb 2026, median of the 20 stack voltages (top) and median of the 36 stack thermocouples (bottom), every 5 min. Grey band: hydrogen briefly off just before hot standby.

Arrive after a stop: wait for hydrogen

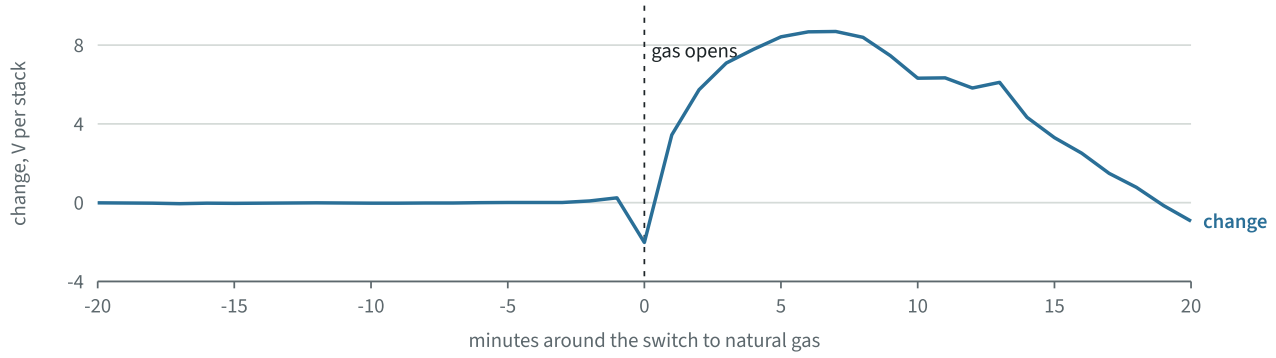
When hot standby follows a stop (an emergency stop or a trip), the hydrogen is often still off for the first minutes. The voltage sits around 80 V per stack until the hydrogen comes back. Then it recovers: most of the way in about 5 minutes, fully in a median of about 14 minutes (longest 42).



24 Mar 2026, 14:26–15:46. Blue: median stack voltage. Gold: hydrogen flow (low = off, high = 60 l/min).

Leave to fuel switch: a jump up

When natural gas is opened, the gas reaching the stack becomes much richer in hydrogen. The resting voltage rises by about **8.7 V per stack** (≈ 35 V per string) within 8 minutes. There is a short dip of ~2 V right at the switch, when the nitrogen and hydrogen are cut and the natural gas has not yet arrived.



Median change in stack voltage around the 12 hot standby → fuel switch changes, relative to the 20 minutes before.

IF YOU REMEMBER ONE THING

In hot standby the voltage is a gas gauge: flat while hydrogen flows, down when it stops, up when natural gas arrives.

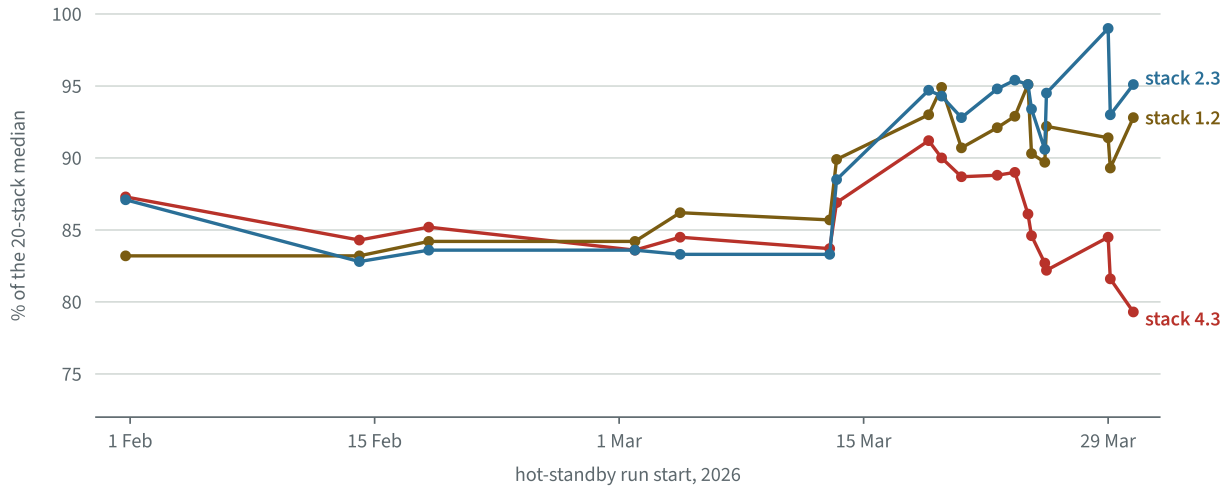
← THE SHORT ANSWER

THE WEAK STACK →

PAGE 3 · THE WEAK STACK

Stack 4.3 rests lower than the others and is getting weaker

To compare stacks fairly, each stack's voltage is shown as a share of the median of all 20 stacks at the same minute. That cancels out temperature and gas, which all stacks share. A healthy stack sits near 100 %.



Each dot is one hot-standby run of at least 30 minutes (first 30 minutes left out). Name "4.3" = string 4, stack 3 (tag EIR 4.4.18); "1.2" = EIR 4.1.15; "2.3" = EIR 4.2.18.

Stack 4.3 — getting worse

About 87 % at the first run (31 Jan), 84–85 % in February, up to 91 % on 18 March, then down to **79–82 %** on 29–30 March. In volts: about 85 V against ~110 V for a typical stack. It is the lowest stack in 52 % of the logged hot-standby minutes. Under load (part load) it also sits low, at 87.5 %.

Stacks 1.2 and 2.3 — recovered

Both started near 83–84 %. From mid-March they rose to 90–95 %. The change came after the plant first ran at ~600 °C and started taking natural gas. Why they improved is not known.

What can make one stack rest lower?

With no current, a stack's resting voltage depends only on the gas on each side of its cells and its temperature. All 20 stacks share the same supply, so a stack that is lower usually means one of three things. These are possible causes, not proven ones:

- **Gas leaking across** a cracked cell or a seal, so fuel and air mix and the voltage drops.
- **Less fuel reaching it**, for example from a partly blocked channel in the manifold.
- **A slightly different temperature** at that stack. The 36 thermocouples are not mapped to stacks, so we cannot check this yet.

A leak tends to get worse over time and with each thermal cycle. That fits the March trend for stack 4.3, but the data alone cannot prove it.

RUN START	MIN	STACK °C	TYPICAL STACK V	4.3 %	1.2 %	2.3 %
01-31 17:47	4814	510	116.8	87.3	83.2	87.1
02-14 02:58	3339	511	111.1	84.3	83.2	82.8
02-18 02:19	907	509	111.4	85.2	84.2	83.6
03-01 21:43	3699	511	111.6	83.6	84.2	83.6
03-04 11:54	362	610	105.9	84.5	86.2	83.3
03-13 01:13	510	578	104.9	83.7	85.7	83.3
03-13 10:42	156	635	108.5	86.9	89.9	88.5
03-18 17:27	1003	584	110.3	91.2	93.0	94.7
03-19 11:14	250	609	112.8	90.0	94.9	94.3
03-20 14:17	127	568	110.8	88.7	90.7	92.8
03-22 15:36	448	577	109.9	88.8	92.1	94.8
03-23 15:54	998	581	110.7	89.0	92.9	95.4
03-24 09:49	237	629	111.5	86.1	95.1	95.1
03-24 14:46	1047	601	109.0	84.6	90.3	93.4
03-25 08:56	75	624	106.8	82.7	89.7	90.6

03-25 11:14	182	584	112.4	82.2	92.2	94.5
03-29 00:00	120	500	106.0	84.5	91.4	99.0
03-29 03:00	1779	580	105.2	81.6	89.3	93.0
03-30 10:40	141	634	108.2	79.3	92.8	95.1

> **IF YOU REMEMBER ONE THING**
> Stack 4.3 has dropped from about 90 % to about 80 % of its neighbours in two weeks of March. It is the one stack to watch.

← THROUGH ONE STANDBY RUN

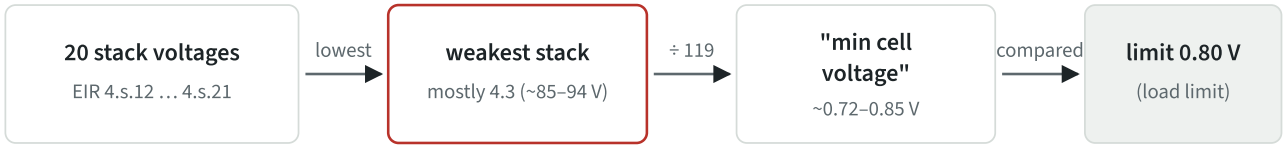
DATA CHECKS AND DECISIONS →

PAGE 4 · DATA CHECKS AND DECISIONS

Three things in the data to know before using it

1 · The "minimum cell voltage" is not a single cell

The plant logs a "minimum cell voltage" for each string. In the data it is always exactly the lowest of the 20 stack voltages divided by **119**, which looks like the number of cells per stack. So it is the average cell of the weakest stack, the same value for all 5 strings.



It is an average over the 119 cells of the weakest stack, not one measured cell

Checked on every minute where both values exist: ratio median 119.00, 25th–75th percentile 118.99–119.00.

IN HOT STANDBY (12,413 MINUTES WITH THE TAG)	SHARE
Below the 0.80 V setting	43 %
Below the 0.75 V control lower limit	17 %
Weakest stack is 4.3 / 1.2 / 2.3	52 / 39 / 10 %

This does no harm in hot standby because no current flows. But when load is switched on, this limit is set by the weakest stack. So stack 4.3 may be the stack that caps the whole plant's power.

2 · Two days of missing stack readings

From 14 to 16 February (about 53 hours), three stack readings show 0 V: stacks 1.1, 1.2 and 2.4. The plant was running normally. Because of this, the string 1 total looks like 226 V in that run instead of ~430 V. These minutes should be marked as bad sensor data, not as a real voltage drop.

3 · A step down between the first two runs

At the same 510 °C, the typical stack rested at about **115–117 V** at the end of the first run (1–3 Feb) and about **111 V** from 14 Feb on, then stayed there. Something changed between the two runs (a cool-down and re-heat lay between). We cannot tell what from the data.

Where the data comes from: per-stack voltages `pi_EIR_4_s_12/15/18/21` (all runs); string totals `TotalVoltage_M1-5` (= sum of the 4 stacks, logged only from 13 Feb); stages from `data_analysis/plant_stages/plant_stages_1min.parquet`.

Decisions for you

	DECISION	SUGGESTION
A	Add a weak-stack check: alarm when one stack rests below a set share of the stack median in hot standby	Warn at 85 %, alert at 80 % (4.3 is at 79–82 % now)
B	Ask the plant team or stack maker whether stack 4.3 has a known leak or fuel-supply issue	Yes, with the page-3 chart
C	Rename the "min cell voltage" tag in our notes and in the digital twin to "weakest-stack average cell voltage"	Yes
D	Mark 14–16 Feb readings of stacks 1.1, 1.2, 2.4 as bad data	Yes
E	Ask what happened between 3 and 14 Feb (step from ~116 to ~111 V)	Ask the plant team

> **IF YOU REMEMBER ONE THING**
> The plant's "minimum cell voltage" is really stack 4.3 divided by 119, so the weak stack is already the one the load limit looks at.

← THE WEAK STACK

BACK TO THE SHORT ANSWER